1.

Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

ANS:-

Blocking Spotify Using Windows Defender Firewall

- Steps:
    1. Open Windows Defender Firewall with Advanced Security.
    2. Select Outbound Rules from the left panel.
    3. Click New Rule and choose Program.
    4. Select the Spotify desktop application (.exe) and click Next.
    5. Choose Block the connection and continue.
    6. Apply the rule to Domain, Private, and Public profiles.
    7. Name the rule Block Spotify and click Finish.
- Result: The outbound firewall rule blocks all network traffic from the Spotify desktop application.

2.

Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

ANS:-

Inbound Firewall Rule for Google Chrome on Port 443 (HTTPS)

Goal: Set up an inbound rule in Windows Defender Firewall to allow incoming connections for Google Chrome on TCP port 443.

- Steps:
    1. Open Windows Defender Firewall with Advanced Security.
    2. Select Inbound Rules and click New Rule.
    3. Choose Port and select TCP.
    4. Enter 443 as the specific local port.
    5. Select Allow the connection.
    6. Apply the rule to the required network profiles.
    7. Name the rule Allow Chrome HTTPS (443) and click Finish.
- Result: The inbound rule allows HTTPS traffic on TCP port 443 for Google Chrome.

3.

Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall. Hint: Use the 'Scope' tab when creating a new rule to specify the IP address.

ANS:-

Blocking Traffic from a Specific IP Address (123.45.67.89)

- Steps:
    1. Open Windows Defender Firewall with Advanced Security.
    2. Select Inbound Rules and click New Rule.
    3. Choose Custom and click Next.
    4. Keep the default settings for Program and Protocol and Ports, then click Next.
    5. In the Scope tab, under Remote IP address, select These IP addresses.
    6. Click Add and enter 123.45.67.89, then click Next.
    7. Select Block the connection and click Next.
    8. Apply the rule to the required network profiles.
    9. Name the rule Block 123.45.67.89 and click Finish.
- Result: All incoming traffic from 123.45.67.89 is blocked by Windows Defender Firewall.

4.

Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.

ANS:-

Comparison Point	**Hardware Firewall**	**Software Firewall**
Location	**A separate physical device placed between the network and the Internet.**	**A program installed and running on an individual computer.**
Protection	**Protects multiple devices across the entire network.**	**Mainly protects the specific device on which it is installed.**
Performance	**Handles network traffic efficiently without consuming individual computer resources**	**Utilizes some of the computer's CPU and memory resources.**
Use Case	**Large Office: Protects the company network from unauthorized Internet traffic.**	**Windows PC: Windows Defender Firewall blocks unwanted connections to protect the individual computer.**